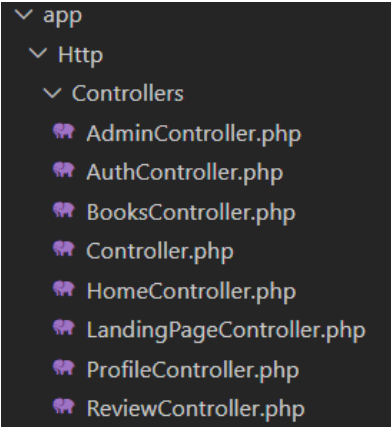


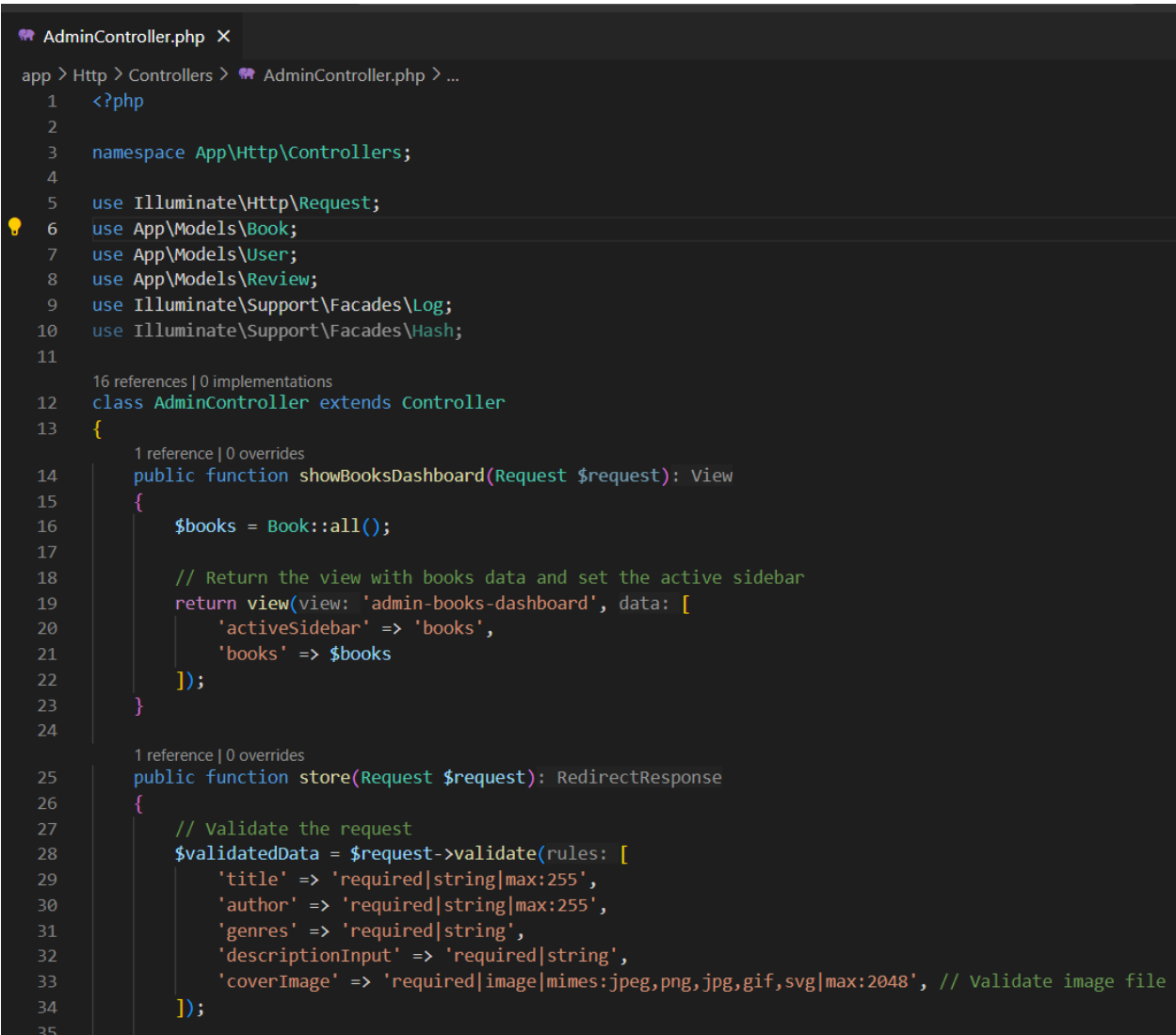
Lab 7: Populating From a Database Documentation

Part 1: Create a database and create a connection to it.

Controllers



AdminController



➔ AdminController handles all admin functions such as managing users, books, and reviews. Admin users can perform actions like deleting users, managing book details, and deleting reviews. It also handles the display of different dashboards for admin users.

BooksController

BooksController.php X

app > Http > Controllers > BooksController.php > ...

```
1 k?php
2
3 namespace App\Http\Controllers;
4
5 use Illuminate\Http\Request;
6 use App\Models\Book;
7 use Illuminate\Support\Facades\Log;
8 use Illuminate\Support\Facades\Auth;
9 use Illuminate\Support\Facades\Hash;
10
11
12 class BooksController extends Controller
13 {
14
15     1 reference | 0 overrides
16     public function show($id): View
17     {
18         $book = Book::find(id: $id);
19
20         if (!$book) {
21             abort(code: 404);
22         }
23
24         // Split the genre string into an array to handle multiple genres
25         $bookGenres = explode(separator: ',', string: $book->genre); // Split the stored genre string into an array
26
27         return view(view: 'books.book-details', data: compact(var_name: 'book', var_names: 'bookGenres'));
28     }
29
30     1 reference | 0 overrides
31     public function showByRating(): View
32     {
33         $booksByRating = Book::withAvg(relation: 'reviews', column: 'rating') // Calculate average rating
34         ->orderByDesc(column: 'reviews_avg_rating') // Order by rating
35         ->get();
```

➔ BooksController manages book-related actions like searching, displaying books by genre and ratings, showing book details, and handling favorite books feature. This controller is also responsible for managing books based on user interactions like genre selection and sorting.

ReviewController

BooksController.php ReviewController.php X

app > Http > Controllers > ReviewController.php > ReviewController > showRatings()

```
14 class ReviewController extends Controller
60 }
61
62
63     1 reference | 0 overrides
64     public function showRatings($id): View
65     {
66         $book = Book::find(id: $id);
67
68         $bookGenre = $book ? $book->genre : 'Unknown';
69
70         $bookGenres = explode(separator: ',', string: $bookGenre); // Split the genres into an array
71
72         $reviews = Review::where(column: 'bookID', operator: $id)
73         ->with(relations: 'replies') // Load replies for each review
74         ->get();
75
76         $ratings = Review::where(column: 'bookID', operator: $id)->pluck(column: 'rating');
77
78         $averageRating = $ratings->avg();
79
80         $totalReviews = $ratings->count();
81
82         $starRatings = [];
83         for ($i = 5; $i >= 1; $i--) {
84             $count = $ratings->filter(callback: fn($rating): bool => $rating == $i)->count();
85             $percentage = $totalReviews > 0 ? ($count / $totalReviews) * 100 : 0;
86             $starRatings[$i] = round(num: $percentage, precision: 1);
87         }
88
89         return view(view: 'books.book-details', data: compact(var_name: 'reviews', var_names: 'starRatings', 'totalReviews'));
90     }
91
92 }
```

➔ ReviewController manages book reviews, allowing users to submit, update, or delete their reviews. It handles the review-related functionality, including adding replies to reviews and managing the review status for books.

AuthController

BooksController.phpAuthController.php X

app > Http > Controllers > AuthController.php > ...

9

5 references | 0 implementations

10class AuthController extends Controller

11{

121 reference | 0 overrides

12public function login(): View

13{

14return view(view: 'auth.login');

15}

16

171 reference | 0 overrides

17function loginPost(Request \$request): mixed|RedirectResponse

18{

19\$request->validate(rules: [

20"email" => 'required',

21"password" => 'required',

22]);

23

24\$credentials = \$request->only(keys: "email", "password");

25

26if (Auth::attempt(credentials: \$credentials)) {

27if (Auth::user()->role === 'admin') {

28return redirect()->route(route: 'admin.books.dashboard');

29}

30

31return redirect()->route(route: 'profile', parameters: ['email' => Auth::user()->email]);

32}

33

34return redirect(to: route(name: "login"))->with(key: "error", value: "Login failed");

35

36

37}

38

391 reference | 0 overrides

39function register(): View

40{

41

42return view(view: 'auth.register');

- ➔ AuthController manages user authentication, including login, registration, and logout processes. It validates user credentials and redirects users based on their authentication status. This controller ensures secure access for both registered and guest users

**I will only explain the important controllers so other controllers will not be mentioned anymore.*

Registering Controllers to Routes

```
Route::get(uri: '/', action: [LandingPageController::class, 'index'])->name(name: 'landing-page');

// Route for searching books (users)
Route::get(uri: '/search', action: [BooksController::class, 'search'])->name(name: 'books.search');

// Authentication Routes (only for guests)
Route::middleware(middleware: 'guest')->group(callback: function (): void {
    Route::get(uri: '/login', action: [AuthController::class, 'login'])->name(name: 'login');
    Route::post(uri: '/login', action: [AuthController::class, 'loginPost'])->name(name: 'login.post');
    Route::get(uri: '/register', action: [AuthController::class, 'register'])->name(name: 'register');
    Route::post(uri: '/register', action: [AuthController::class, 'registerPost'])->name(name: 'register.post');
});
```

- ➔ The AuthController is responsible for user authentication, including login and registration. It is wrapped in the guest middleware, ensuring that only guests (unauthenticated users) can access the login and registration routes.

```
// Authenticated Routes (for logged-in users)
Route::middleware(middleware: 'auth')->group(callback: function (): void {
    Route::get(uri: '/contact', action: function (): View {
        return view(view: 'contact');
    });

    Route::get(uri: '/home', action: [HomeController::class, 'index'])->name(name: 'home.index');
    Route::get(uri: '/reviews', action: [ReviewController::class, 'show']);

    Route::post(uri: '/logout', action: function (): RedirectResponse {
        auth()->logout(); // Logs out the user
        return redirect(to: route(name: 'landing-page'));
    })->name(name: 'logout');

    // Content loading routes
    Route::get(uri: '/home/content', action: [BooksController::class, 'loadBooks'])->name(name: 'home.books');
```

```
Route::get(uri: '/browse', action: [BooksController::class, 'showByGenre'])->name(name: 'books.byGenre');
//will trigger the showByGenre method ng BookController, yung showByGenre will fetch all the genres first (parang naka group by para one instance)
//after that it will pass variable sa method niya tas yung unique genres nasa taas parang category (referring sa browse view)
//then each genre naka loop siya will display the books related dun sa genre (limit 4)

Route::get(uri: '/browse-books/{genre}', action: [BooksController::class, 'showGenre'])->name(name: 'books.browse.genre');
//yung category sa browse view pag clinick yung genre it will show all the books to that genre (view rendered is show-books-by-genre)

Route::get(uri: '/browse-books/view-all/{genre}', action: [BooksController::class, 'viewAllByGenre'])->name(name: 'view-all.genre.show');

Route::get(uri: '/browse/books/view-all/release-date/{genre}', action: [BooksController::class, 'showbyReleaseDate'])->name(name: 'view-all.latest');
// nakabased to sa released at column sa books since latest books siya, see logic sa method niya

Route::get(uri: '/browse/books/view-all/rating/{genre}', action: [BooksController::class, 'showByRating'])->name(name: 'view-all.rating.show');

Route::get(uri: '/browse-books/book-detail/{id}', action: [BooksController::class, 'showBookDetail'])->name(name: 'books.bookDetail');

Route::post(uri: '/browse-books/{id}/reply/{review_id}', action: [ReviewController::class, 'addReply'])->name(name: 'reviews.reply');

Route::post(uri: '/book/{id}/review', action: [ReviewController::class, 'store'])->name(name: 'reviews.store');

Route::put(uri: '/reviews/{reviewID}/update', action: [ReviewController::class, 'update'])->name(name: 'reviews.update');

Route::delete(uri: '/reviews/{reviewID}', action: [ReviewController::class, 'delete'])->name(name: 'reviews.delete');

Route::get(uri: '/profile', action: [ProfileController::class, 'showProfile'])->name(name: 'profile');

Route::post(uri: '/favorite/{bookId}/toggle', action: [BooksController::class, 'toggleFavorite'])->name(name: 'favorite.toggle');
```

- ➔ BooksController is responsible for managing book-related routes.
- ➔ ReviewController handles the posting, updating, and deleting of reviews. Like BooksController, its routes are wrapped in the auth middleware, requiring users to be authenticated before interacting with reviews.

```
// Authentication Routes (only for admin)
Route::middleware(middleware: ['auth', 'admin'])->group(callback: function (): void {
    // Show dashboard views
    Route::get(uri: '/admin-books-dashboard', action: [AdminController::class, 'showBooksDashboard'])->name(name: 'admin.books.dashboard');
    Route::get(uri: '/admin/admin-users-dashboard', action: [AdminController::class, 'showUsersDashboard'])->name(name: 'admin.users.dashboard');
    Route::get(uri: '/admin/admin-reviews-dashboard', action: [AdminController::class, 'showReviewsDashboard'])->name(name: 'admin.reviews.dashboard');

    // Routes for Admin User Management
    Route::get(uri: '/admin/users/search', action: [AdminController::class, 'adminUserSearch'])->name(name: 'admin.users.search');
    Route::delete(uri: '/admin/users/delete/{id}', action: [AdminController::class, 'deleteUser'])->name(name: 'admin.users.delete');

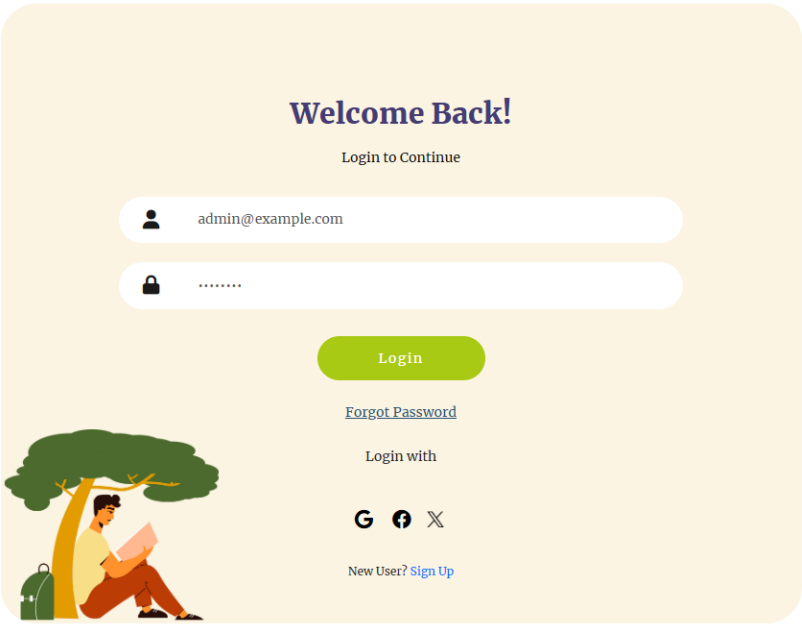
    //Routes for Admin Review Management
    Route::get(uri: '/admin/reviews/search', action: [AdminController::class, 'adminReviewSearch'])->name(name: 'admin.reviews.search');
    Route::post(uri: '/admin/reviews/delete', action: [AdminController::class, 'adminDeleteReviews'])->name(name: 'admin.reviews.delete');
    Route::post(uri: '/admin/review/{userID}/violation', action: [AdminController::class, 'incrementViolation'])->name(name: 'admin.reviews.violation');

    //Routes for Admin Book Management
    Route::get(uri: '/admin/books/search', action: [AdminController::class, 'adminBookSearch'])->name(name: 'admin.books.search'); // Route for search
    Route::get(uri: '/admin/books/search/genre', action: [AdminController::class, 'adminSearchByGenre'])->name(name: 'adminSearchByGenre'); // Route to show the edit form (AJAX)
    Route::get(uri: '/books/edit/{bookID}', action: [AdminController::class, 'edit'])->name(name: 'books.edit'); // Route to show the edit form (AJAX)
    Route::put(uri: '/books/update/{book}', action: [AdminController::class, 'update'])->name(name: 'books.update'); // Route to update the book details
    Route::post(uri: '/books/add', action: [AdminController::class, 'store'])->name(name: 'books.add'); // Route to add a book
    Route::get(uri: '/books/add', action: [AdminController::class, 'addBookSuccess'])->name(name: 'books.index'); // Route to show success prompt
    Route::post(uri: '/books/delete', action: [AdminController::class, 'deleteBooks'])->name(name: 'books.delete'); // Route to delete a book
});
```

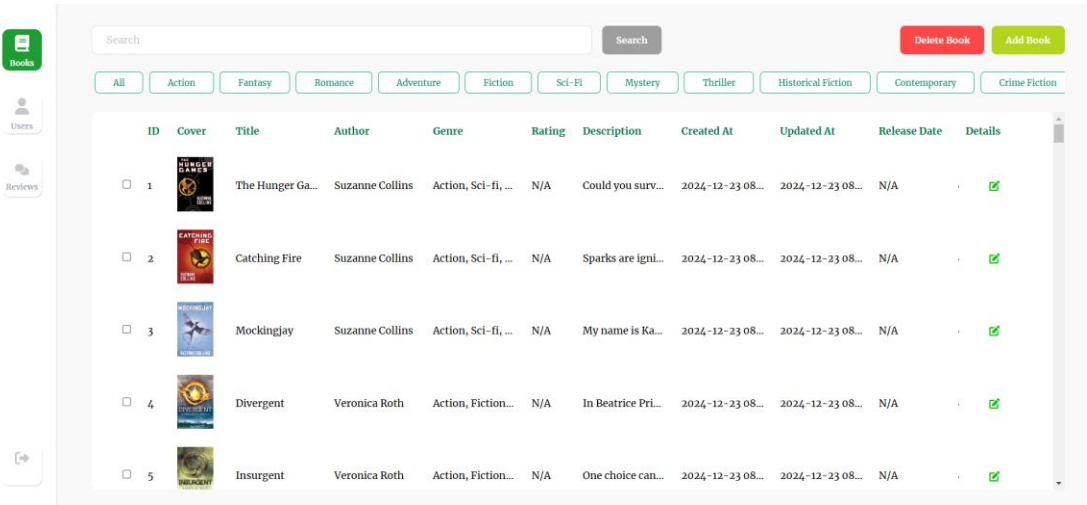
- ➔ AdminController is wrapped in both the auth and admin middleware, ensuring that only authenticated users with admin privileges can access the admin routes.

Part 2: Make an account registration and login page.

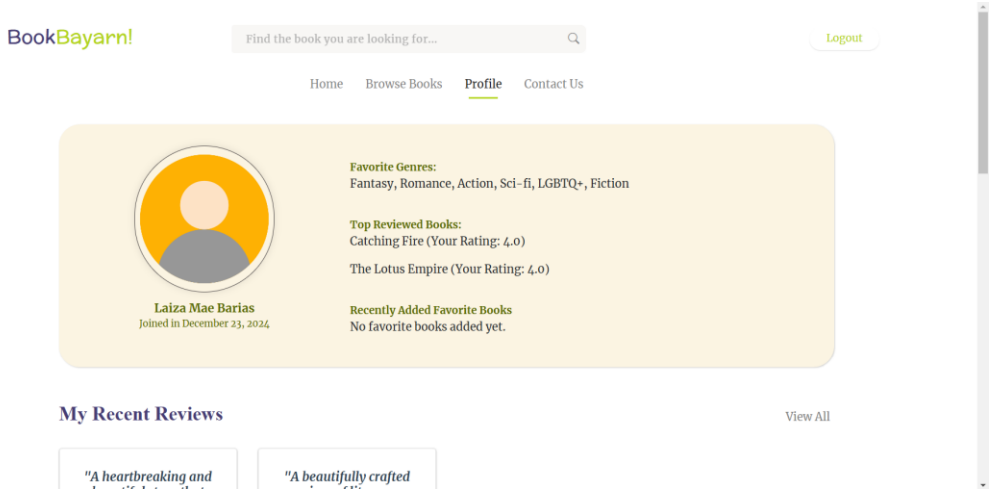
For Admin



➔ The admin has a set email and password and once logged in, they will be redirected to the admin books dashboard



➔ While authenticated users will be directed to their profile page.



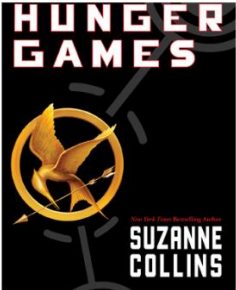
Part 3: Populate your pages using data fetched from a database.

Books Table in the Database

		bookID	title	author	genre	description	cover	created_at	updated_at	release_date
		1	The Hunger Games	Suzanne Collins	Action, Sci-fi, Fantasy	Could you survive on your own in the wild, with ev...	assets/covers/The Hunger Games.jpg	2024-12-23 06:39:02	2024-12-23 06:39:02	NULL
		2	Catching Fire	Suzanne Collins	Action, Sci-fi, Fantasy, Romance	Sparks are igniting. Flames are spreading. And the...	assets/covers/Catching Fire.jpg	2024-12-23 06:39:02	2024-12-23 06:39:02	NULL
		3	Mockingjay	Suzanne Collins	Action, Sci-fi, Fantasy, Adventure, Fiction, Roman...	My name is Katniss Everdeen. Why am I not dead? I ...	assets/covers/Mockingjay.jpg	2024-12-23 06:39:02	2024-12-23 06:39:02	NULL
		4	Divergent	Veronica Roth	Action, Fiction, Fantasy, Sci-Fi, Romance	In Beatrice Prior's dystopian Chicago world, socie...	assets/covers/Divergent.jpg	2024-12-23 06:39:02	2024-12-23 06:39:02	NULL
		5	Insurgent	Veronica Roth	Action, Fiction, Fantasy, Sci-Fi, Romance	One choice can transform you—or it can destroy you...	assets/covers/Insurgent.jpg	2024-12-23 06:39:02	2024-12-23 06:39:02	NULL
		6	Allegiant	Veronica Roth	Action, Fiction, Fantasy, Sci-Fi, Romance	The faction-based society that Tris Prior once bel...	assets/covers/Allegiant.jpg	2024-12-23 06:39:02	2024-12-23 06:39:02	NULL
		7	The Maze Runner	James Dashner	Action, Fiction, Fantasy, Sci-Fi, Thriller	If you ain't scared, you ain't human. When Thomas ...	assets/covers/The Maze Runner.jpg	2024-12-23 06:39:02	2024-12-23 06:39:02	NULL
		8	Maze Runner: The Scorch Trials	James Dashner	Action, Fiction, Fantasy, Sci-Fi, Thriller	Escaping the Maze was only the beginning. The worl...	assets/covers/Scorch Trials.jpg	2024-12-23 06:39:02	2024-12-23 06:39:02	NULL
		9	Maze Runner: The Death Cure	James Dashner	Action, Fiction, Fantasy, Sci-Fi, Thriller	An all new, original Maze Runner story set directl...	assets/covers/Death Cure.jpg	2024-12-23 06:39:02	2024-12-23 06:39:02	NULL
		10	The Lightning Thief	Rick Riordan	Action, Adventure,	Percy Jackson is a good kid,	assets/covers/The Lightning Thief.jpg	2024-12-23	2024-12-23	NULL

Home Page

Top Rated & Popular Now



The Hunger Games

Suzanne Collins

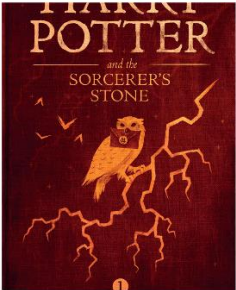
5.0 ★★★★★



Dungeon Crawler Carl

Matt Dinniman

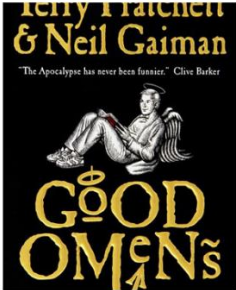
5.0 ★★★★★



Harry Potter and the Sorcerer's Stone

J.K. Rowling

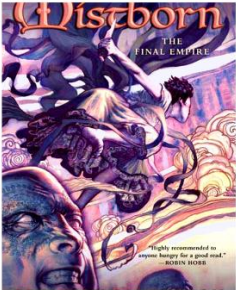
5.0 ★★★★★



Good Omens

Neil Gaiman and Terry Pratchett

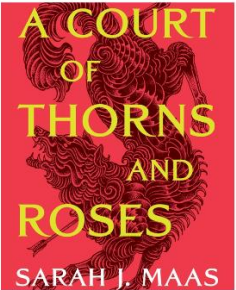
5.0 ★★★★★



Mistborn: The Final Empire

Brandon Sanderson

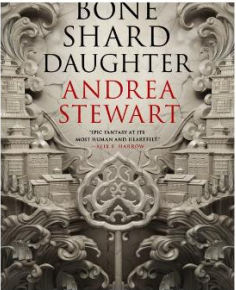
5.0 ★★★★★



A Court of Thorns and Roses

Sarah J. Maas

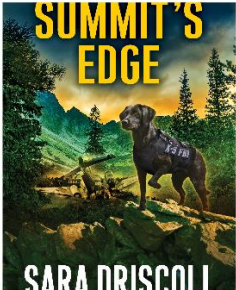
5.0 ★★★★★



The Bone Shard Daughter

Andrea Stewart

5.0 ★★★★★



Summit's Edge

Sara Driscoll

4.7 ★★★★★☆



Maze Runner: The Death Cure



The Lightning Thief

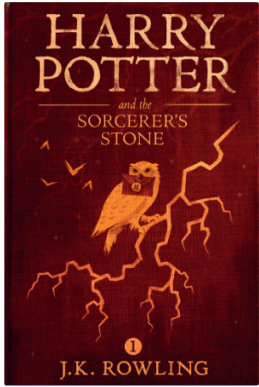


The Maze Runner



Good Omens

➔ This shows cards of books populated by book details from our database based on ratings.



Add to Favorites

Rate this Book

Harry Potter and the Sorcerer’s Stone

by J.K. Rowling

Synopsis:

Turning the envelope over, his hand trembling, Harry saw a purple wax seal bearing a coat of arms; a lion, an eagle, a badger and a snake surrounding a large letter 'H'.

Harry Potter has never even heard of Hogwarts when the letters start dropping on the doormat at number four, Privet Drive. Addressed in green ink on yellowish parchment with a purple seal, they are swiftly confiscated by his grisly aunt and uncle. Then, on Harry's eleventh birthday, a great beetle-eyed giant of a man called Rubeus Hagrid bursts in with some astonishing news: Harry Potter is a wizard, and he has a place at Hogwarts School of Witchcraft and Wizardry. An incredible adventure is about to begin!

Genres: Children’s Adventure Sci-fi Fantasy

➔ If a book is clicked, it will show the specific details of that book such as the title, author, cover image, genres, and description.

Ratings & Reviews

Total Reviews: 2



LightRiver ★ 5

A brilliant and emotional memoir. The humor in this book will make you laugh out loud, but the poignant moments will leave you thinking long after.

Published on December 23, 2024

➔ Below it shows the ratings and reviews.

LightRiver ★ 5

A brilliant and emotional memoir. The humor in this book will make you laugh out loud, but the poignant moments will leave you thinking long after.

Published on December 23, 2024

Laiza Mae Barias

YES! I love the trio <3

Published on December 23, 2024

➔ Users can reply to comments.

Add a Review

Your Rating: 5 Stars

I love this book!

Submit Review

Laiza Mae Barias ★ 5

I love this book!

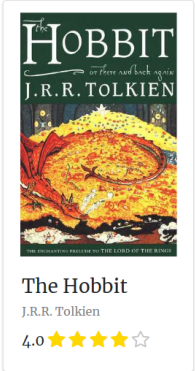
Published on December 23, 2024

Edit

Delete

➔ Users can also add, edit, and delete a review.

You might also like



➔ And below shows recommendations based on the genre.

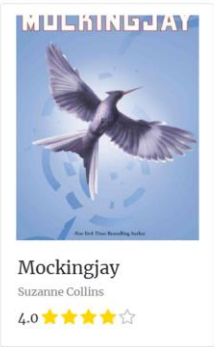
Search Feature

Search Results for "catching fire"



➔ Users can search books based on title, author, or genre.

Search Results for "suzanne"



Profile

Home

Browse Books

Profile

Contact Us

Laiza Mae Barias

Joined in December 23, 2024

Favorite Genres:

Fantasy, Romance, Action, Sci-fi, LGBTQ+, Fiction

Top Reviewed Books:

Catching Fire (Your Rating: 4.0)

The Lotus Empire (Your Rating: 4.0)

Recently Added Favorite Books

No favorite books added yet.

My Recent Reviews

[View All](#)

"A heartbreaking and beautiful story that examines the fragility of life and the power of love. It's a deeply emotional read."

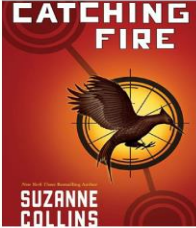
Catching Fire

"A beautifully crafted piece of literary fiction. The author's prose is lyrical, and the story is filled with emotion and depth."

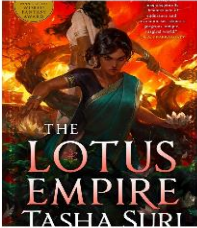
The Lotus Empire

My Reviewed Books

[View All](#)



Catching Fire
Suzanne Collins
4.3 ★★★★★☆



The Lotus Empire
Tasha Suri
4.5 ★★★★★☆

➔ Profile is populated with users data such as name, date they joined (created their account), favourite genres based on reviews, top reviewed books, and recently added favourite books.

** All information displayed was queried from the database and dynamically rendered on the webpage, same with the tables below.*

Books

Users

Reviews

Search

Delete Book

Add Book

All

Action

Fantasy

Romance

Adventure

Fiction

Sci-Fi

Mystery

Thriller

Historical Fiction

Contemporary

Crime Fiction

	ID	Cover	Title	Author	Genre	Rating	Description	Created At	Updated At	Release Date	Details
☐	1		The Hunger Ga...	Suzanne Collins	Action, Sci-fi, ...	N/A	Could you surv...	2024-12-23 08:...	2024-12-23 08:...	N/A	
☐	2		Catching Fire	Suzanne Collins	Action, Sci-fi, ...	N/A	Sparks are igni...	2024-12-23 08:...	2024-12-23 08:...	N/A	
☐	3		Mockingjay	Suzanne Collins	Action, Sci-fi, ...	N/A	My name is Ka...	2024-12-23 08:...	2024-12-23 08:...	N/A	
☐	4		Divergent	Veronica Roth	Action, Fiction...	N/A	In Beatrice Pri...	2024-12-23 08:...	2024-12-23 08:...	N/A	
☐	5		Insurgent	Veronica Roth	Action, Fiction...	N/A	One choice can...	2024-12-23 08:...	2024-12-23 08:...	N/A	

Books

Users

Reviews

Search Clear

UserID	Name	Email	Joined At	Updated At	Violations	Action
1	admin	admin@example.com	2024-12-23 08:07:43	2024-12-23 08:07:43	-	-
2	The Ugly King	uglyking@example.com	2024-12-23 08:07:43	2024-12-23 08:07:43	-	-
3	Silent Shadow	silentshadow@example.com	2024-12-23 08:07:43	2024-12-23 08:07:43	-	-
4	Lost Pirate	lostopirate@example.com	2024-12-23 08:07:43	2024-12-23 08:07:43	-	-
5	BlueSky	bluesky@example.com	2024-12-23 08:07:43	2024-12-23 08:07:43	-	-
6	RedApple	redapple@example.com	2024-12-23 08:07:43	2024-12-23 08:07:43	-	-
7	GreenLeaf	greenleaf@example.com	2024-12-23 08:07:43	2024-12-23 08:07:43	-	-
8	HappyDay	happyday@example.com	2024-12-23 08:07:43	2024-12-23 08:07:43	-	-
9	QuietStar	quietstar@example.com	2024-12-23 08:07:43	2024-12-23 08:07:43	-	-
10	BrightSun	brightsun@example.com	2024-12-23 08:07:43	2024-12-23 08:07:43	-	-
11	SoftCloud	softcloud@example.com	2024-12-23 08:07:43	2024-12-23 08:07:43	-	-
12	SilverMoon	silvermoon@example.com	2024-12-23 08:07:43	2024-12-23 08:07:43	-	-
13	QuickFox	quickfox@example.com	2024-12-23 08:07:43	2024-12-23 08:07:43	-	-
14	LazyBear	lazybear@example.com	2024-12-23 08:07:43	2024-12-23 08:07:43	-	-

Books

Users

Reviews

Search

Search

Clear

Delete Review

Show Negative Comments

All

1 Star

2 Stars

3 Stars

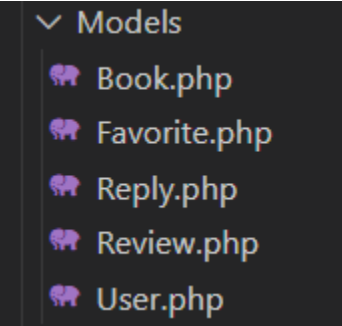
4 Stars

5 Stars

	ID	Book Title	User	Rating	Comment	Date Created	Date Updated	Action
☐	1	The Hunger Games	The Ugly King	5.0	A gripping dystopian novel that keeps you on the edge of your seat. Katniss's journey is both heart-wrenching and inspiring.	2024-12-23 08:07:43	2024-12-23 08:07:43	-
☐	2	Catching Fire	Silent Shadow	4.0	The sequel to The Hunger Games surpasses the first book with more intense action and deeper character development.	2024-12-23 08:07:43	2024-12-23 08:07:43	-
☐	3	Mockingjay	Lost Pirate	4.0	A powerful conclusion to the trilogy that explores the complexities of war and sacrifice, though it leaves you with mixed emotions.	2024-12-23 08:07:43	2024-12-23 08:07:43	-
☐	4	Divergent	BlueSky	4.0	An intriguing world with a strong, independent heroine, though the pacing slows down at times.	2024-12-23 08:07:43	2024-12-23 08:07:43	-
☐	5	Insurgent	RedApple	4.0	A thrilling continuation with unexpected twists and deeper world-building. The stakes are raised higher.	2024-12-23 08:07:43	2024-12-23 08:07:43	-
☐	6	Allegiant	GreenLeaf	3.0	An emotional rollercoaster that ties up the series, though the ending is polarizing for many readers.	2024-12-23 08:07:43	2024-12-23 08:07:43	-
☐	7	The Maze Runner	HappyDay	4.0	A fast-paced thriller with mystery and action that keeps you guessing until the very end.	2024-12-23 08:07:43	2024-12-23 08:07:43	-
☐	8	Maze Runner: The Scorch Trials	QuietStar	4.0	A worthy follow-up to The Maze Runner. The Scorch Trials add even more danger and excitement to the series.	2024-12-23 08:07:43	2024-12-23 08:07:43	-
☐	9	Maze Runner: The Death Cure	BrightSun	4.0	A satisfying conclusion to the Maze Runner series that ties up all loose ends, though it can be a bit predictable at times.	2024-12-23 08:07:43	2024-12-23 08:07:43	-
☐	10	The Lightning Thief	SoftCloud	4.0	A fun, adventurous read with likable characters. The plot keeps moving at a	2024-12-23	2024-12-23	-

Models

Book Model



➔ These are the models we have for each table.

Book.php

app > Models > Book.php > Book

```
4
5 use Illuminate\Database\Eloquent\Model;
6 use Illuminate\Database\Eloquent\Factories\HasFactory;
7
8 class Book extends Model
9 {
10     use HasFactory;
11
12     protected $table = 'books';
13
14     public $primaryKey = 'bookID';
15
16     public $timestamps = true;
17
18     protected $fillable = ['title', 'author', 'genre', 'description', 'cover'];
19
20     // Define the one-to-many relationship with Review
21     public function reviews(): HasMany
22     {
23         return $this->hasMany(related: Review::class, foreignKey: 'bookID'); // The 'bookID' is the foreign
24     }
25
26
27     public function usersWhoFavorited(): BelongsToMany
28     {
29         return $this->belongsToMany(related: User::class, table: 'favorites', foreignPivotKey: 'book_id', re
30     }
31
32
```

➔ The Book model represents the books table and has relationships with the Review and User models. It allows each book to have many reviews and be marked as a favorite by multiple users. The fillable property ensures only certain fields can be mass-assigned.

Reply Model

```
Reply.php X
app > Models > Reply.php > ...
3 namespace App\Models;
4
5 use Illuminate\Database\Eloquent\Factories\HasFactory;
6 use Illuminate\Database\Eloquent\Model;
7
8 class Reply extends Model
9 {
10     use HasFactory;
11
12     // Define the table name (optional if you follow Laravel's conventions)
13     protected $table = 'replies';
14
15     // Mass assignable attributes
16     protected $fillable = [
17         'review_id', // Foreign key to the review
18         'user_id',   // Foreign key to the user
19         'comment',   // Reply content
20     ];
21
22     // A reply belongs to a review
23
24     public function review(): BelongsTo
25     {
26         return $this->belongsTo(related: Review::class, foreignKey: 'review_id');
27     }
28
29     // A reply belongs to a user
30
31     public function user(): BelongsTo
32     {
33         return $this->belongsTo(related: User::class);
34     }
35 }
```

- ➔ Reply model represents the replies table and defines relationships with the Review and User models. Each reply belongs to a specific review and user, which are set using foreign keys (review_id and user_id). The fillable property ensures that only certain fields can be mass-assigned, such as review_id, user_id, and comment.

Review Model

```
Review.php
app > Models > Review.php > ...
1  <?php
2
3  namespace App\Models;
4
5  use Illuminate\Database\Eloquent\Model;
6  use Illuminate\Database\Eloquent\Factories\HasFactory;
7
8  18 references | 0 implementations
9  class Review extends Model
10 {
11     use HasFactory;
12
13     0 references
14     protected $table = 'reviews';
15
16     0 references
17     protected $primaryKey = 'reviewID';
18
19     0 references
20     public $timestamps = true;
21
22     // Define the inverse of the one-to-many relationship with Book
23     0 references
24     protected $fillable = ['userID', 'bookID', 'rating', 'comment'];
25
26     // Define the relationship for replies
27
28     0 references | 0 overrides
29     public function user(): BelongsTo
30     {
31         return $this->belongsTo(related: User::class, foreignKey: 'userID');
32     }
33
34     0 references | 0 overrides
35     public function replies(): HasMany
36     {
37         return $this->hasMany(related: Reply::class, foreignKey: 'review_id'); // 'review_id' is the foreign key in the replies table
38     }
39
40     0 references | 0 overrides
41     public function book(): BelongsTo
42     {
43         return $this->belongsTo(related: Book::class, foreignKey: 'bookID');
44     }
45
46     0 references | 0 overrides
47     public function reviews(): HasMany
48     {
49         return $this->hasMany(related: Review::class, foreignKey: 'bookID');
50     }
51 }
```

- ➔ Review model represents the reviews table and establishes relationships with the User, Book, and Reply models. Each review is associated with a user (through userID), a book (through bookID), and can have multiple replies. The fillable property allows mass assignment for the userID, bookID, rating, and comment fields, while the relationships are defined using Eloquent methods (belongsTo and hasMany).

User Model

```
User.php x
app > Models > User.php > ...
10  class User extends Authenticatable

28      * The attributes that should be hidden for serialization.
29      *
30      * @var array<int, string>
31      */
32      0 references
33      protected $hidden = [
34          'password',
35          'remember_token',
36      ];
37
38      /**
39      * Get the attributes that should be cast.
40      * @return array<string, string>
41      */
42      0 references | 0 overrides
43      protected function casts(): array
44      {
45          return [
46              'email_verified_at' => 'datetime',
47              'password' => 'hashed',
48          ];
49      }
50      0 references | 0 overrides
51      public function reviews(): HasMany
52      {
53          return $this->hasMany(related: Review::class, foreignKey: 'userID');
54      }
55
56      0 references | 0 overrides
57      public function favoriteBooks(): BelongsToMany
58      {
59          return $this->belongsToMany(related: Book::class, table: 'favorites', foreignPivotKey: 'user_id', re
```

- ➔ User model represents a user in the system, with properties like name, email, and password being mass assignable. It defines relationships with the Review model (one-to-many) and the Book model (many-to-many through the favorites pivot table). Additionally, sensitive attributes like password and remember_token are hidden during serialization, and the casts method ensures proper data handling for email_verified_at and password.



** Merry Christmas sir!*